

Chapter 1: From Cosmos to Machine

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From Cosmos to Machine: How a Metaphor Became a Worldview

For most of human history, the world was alive.

Not metaphorically. Not as a comforting fiction told by people who didn't know any better. The cosmos was understood as intelligent, relational, saturated with meaning — and this was the considered view of the most rigorous thinkers in every major civilization. Plato saw the cosmos as a living creature endowed with soul and intellect. The Taoist tradition understood reality as a dynamic flow of complementary forces. In India, Advaita Vedanta recognized a fundamental identity between individual consciousness and the cosmic ground. Indigenous cultures worldwide experienced the world through participatory knowing — a direct relationship with a living environment where interiority was not a private human accident but a pervasive feature of the landscape.

These traditions disagreed about nearly everything else. But they converged on a structural recognition: that mind and matter, interiority and exteriority, are woven together at the deepest level of what is real. Owen Barfield called this *original participation* — a mode of consciousness in which humans experienced themselves as participating in the life of nature rather than standing apart from it.

The modern scientific worldview replaced this understanding with something else: a universe of dead matter in mindless motion, with consciousness as a strange and possibly illusory afterthought. That replacement felt like discovery. But it was actually a choice made for specific reasons, a reorientation that could have gone differently and that we can still reconsider.

That choice matters now because it shapes what we can recognize. In 2018, an orca named Tahlequah carried her dead calf through the Salish Sea for seventeen days. Millions of people saw grief. The scientific worldview told them they were projecting. But the authority of that worldview doesn't come from having settled the question of animal interiority. It comes from four centuries of institutional momentum behind a metaphor that treats the universe as a machine.

The cosmologist Edward Harrison offered a concept that will serve throughout this essay. Every civilization, he observed, constructs what he called a *universe* — a comprehensive model of reality, experienced by those inside it not as a model but as the world itself. These universes are not the Universe. They are masks, fitted to the face of an unknown and perhaps unknowable underlying reality. The living world described in this chapter is one such mask. The mechanical universe that displaced it is another. Masks are replaced not because they fail entirely but because their exclusions eventually become too costly to ignore.

The Choice

The shift began before Galileo, but his move was decisive. He argued that the world is fundamentally accessible only through mathematics — that the “book of nature is written in geometrical characters.” This was not merely a practical decision about which tools to use. It was a reorientation that reshaped Western thought for four centuries.

By focusing exclusively on what could be quantified, Galileo created an implicit hierarchy. Quantities — mass, velocity, position — became “primary qualities” because they could be captured mathematically. Qualities like color, warmth, taste, purpose, and meaning became “secondary” because they resisted mathematical

description. As the mechanistic worldview proved its spectacular predictive power, these qualitative dimensions were gradually deemed less real, less worthy of systematic attention, and eventually came to be seen not as part of nature itself but as mere projections of the perceiving mind.

What began as *let's study what we can measure* carried within it the seeds of *only what we can measure is real*. A pragmatic methodology had become a metaphysical commitment.

The actual history was messier than this clean narrative suggests — Newton was as deeply engaged with alchemy as with physics, and the founders of modern science saw no conflict between mechanism and faith. But the drift was gradual, so each generation could reasonably see themselves as simply building on their predecessors' work. Newton saw universal gravitation as proof of God's rational design — yet the elegance of his mathematical system made the metaphor of a soul-less machine increasingly plausible. By the time Laplace presented his “clockwork universe” to Napoleon and famously replied that he “had no need of that hypothesis,” the transformation was nearly complete.

Alfred North Whitehead called the result the “bifurcation of nature” — the decisive cleaving of what had been experienced as a seamless, meaningful whole. In roughly three centuries — a blink in human history — Western culture traded a view where inner meaning and outer form were co-fundamental for one where only the outer, physical form was considered real. Eventually, many of our best minds came to believe that the *Logos* of the Stoics, the *Tao* of the sages, and the living world of our ancestors were merely creations of the human mind — derivatives of electrochemical processes in our brains.

This 300-year experiment in physicalism is the exception, however, not the rule. When we struggle to fit consciousness into the physicalist framework, we are not struggling with nature. We are struggling with the limitations of a specific, historically contingent metaphor.

The Invisible Architecture of Thought

How did a methodological choice harden into a claim about what is real? Through something deeper than explicit argument: the metaphors through which we think.

Cognitive linguist George Lakoff and philosopher Mark Johnson established the evidence for this. Metaphors are not merely ways of talking about reality — they are ways of *thinking about* reality. We don't just happen to speak of arguments in terms of war (“defending a position,” “attacking weak points”); we actually think about arguments through the structure of combat. These aren't conscious choices. They are cognitive infrastructure — largely invisible frameworks that shape what we can perceive, what questions we can ask, and what answers seem reasonable.

The metaphor of the universe as a machine determines what we can notice. Machines have parts but not purposes. They can be disassembled and optimized but not participated in. They operate according to deterministic laws but lack interiority, meaning, or value. The metaphor makes certain questions natural — *How does it work? What are its mechanisms?* — while rendering others nonsensical: *What does it mean? What is its purpose? What is it like to be this?*

Jeremy Lent has extended this insight to civilizational scale. The root metaphors animating a culture — its deepest assumptions about what reality is — shape everything from social organization to ecological relationship to conceptions of human flourishing. A culture's choice of root metaphor is never merely intellectual. It cascades into institutions, ethics, technology, and the felt quality of daily life.

This is where the argument becomes personal. The metaphor we adopt determines whether Tahlequah and her pod are participants in a living reality or merely complex biological machines. That determination shapes whether, or how urgently, we act to prevent their extinction. When we watched her carry her dead calf for seventeen days, the mechanistic framework told us our emotional recognition was methodologically suspect — that we were projecting. But the framework that made such recognition seem naive is itself a metaphor, not a finding.

The Wrong Template

The machine metaphor didn't just reshape how we think about nature. It reshaped how we evaluate knowledge itself.

The spectacular success of physics in predicting planetary motions created an implicit standard: real science produces mathematical laws that enable precise prediction. This became the template against which all other inquiry would be judged. But it fits some domains far better than others.

Physics succeeds brilliantly with closed, reversible systems with few variables — pendulums, planets, particles in controlled conditions. Biology operates differently. Living systems are open — constantly exchanging energy and matter with their environments. They are irreversible — each moment represents genuinely novel emergence. And they exhibit goal-directedness that resists mechanical description. Evolutionary science compels not because it generates physics-style predictions but because independent lines of evidence — fossil sequences, radiometric dating, comparative genomics, biogeography — converge on the same broad picture. It is the coherence across evidence, not predictive precision, that makes the account compelling.

If biology succeeds through consilience rather than prediction, then consciousness — the ultimate interiority, the paradigm case of irreversible, context-dependent emergence — cannot reasonably be held to standards designed for reversible, isolated, purely exterior systems. The question should not be whether consciousness-inclusive frameworks can meet physics standards, but whether they offer a more coherent account of the full range of evidence than the emergence stories that dominate current thinking.

There is a deeper issue here. Science won its cultural authority not through explanation in any deep philosophical sense, but through prediction and technological control. Newton famously declined to explain gravity — *hypotheses non fingo* — while describing its behavior with extraordinary precision. When that prestige gets extended to consciousness as a promissory note — “we haven't explained it yet, but we will” — it is borrowing credibility earned in a different domain. Qualitative-inclusive frameworks are not failing to meet a neutral standard. They are being evaluated by a criterion constructed within the framework they are questioning.

The Lock-In

Even as these limitations became apparent, Western culture chose not to recognize them but to double down through institutional restructuring.

The 18th century saw the beginning of a division between sciences and humanities that would accelerate over the next two centuries. What had been integrated domains of inquiry — natural philosophy encompassing both physical study and questions of meaning — were systematically cleaved apart. By the mid-20th century, universities had fully organized into separate colleges, funding agencies prioritized quantifiable research, and “scientific” became synonymous with “legitimate” across vast domains of culture. When C.P. Snow delivered his “Two Cultures” lecture in 1959, he wasn't lamenting the division. He was arguing that technological knowledge should dominate education and policy.

The mechanistic worldview became self-perpetuating — not because it had proven adequate to explain consciousness or meaning, but because institutions had been restructured to favor only the kinds of questions it could answer. The framework stopped needing to justify itself. It became the background against which all other claims are evaluated.

Which brings us back to Tahlequah. When we watched her push her calf through the waves, we weren't just seeing interesting behavior. We were encountering what this institutional framework has trained us not to recognize: consciousness that matters, grief that's real, interiority that demands ethical response.

Beyond the Machine

The mechanistic model was historically contingent, built on choices about which aspects of reality to privilege. Those choices generated immense technological power — but also systematic blindness to consciousness, meaning, and participation. We cannot return to premodern frameworks. But we can move beyond the

machine metaphor toward a more complete understanding — one that preserves scientific rigor while recovering what a mechanical lens systematically excludes.

This is not regression to pre-scientific mythology. It is progression toward a more complete picture. The most advanced frontiers of modern physics are beginning to echo the oldest intuitions of the traditions we set aside. Physicist David Bohm proposed an “Implicate Order” — a deeper reality from which both mind and matter emerge — that bears striking structural resemblance to the Neoplatonic “One” and the Buddhist concept of dependent origination. The dual-aspect monism developed by physicist Wolfgang Pauli and psychologist Carl Jung treats the mental and the physical as two ways of perceiving a single, underlying reality. The ancient historical norm may align better with 21st-century science than the 19th-century mechanical model ever did.

It is worth being precise about what kind of physicalism this essay challenges. Not every thinker working within a broadly physicalist framework holds the stark position that consciousness is merely a byproduct awaiting neural explanation. The target is what we might call *default physicalism*: the unexamined assumption, embedded in most scientific and educational institutions, that consciousness is derivative, late, and ultimately explicable without remainder by physical mechanism. The more sophisticated revisions of physicalism, as we’ll see, tend to converge on conclusions practically indistinguishable from the qualitative-inclusive frameworks this essay develops — which is itself a significant finding.

The next chapter examines why that default physicalist framework faces challenges it cannot resolve on its own terms.

This is a summary of Chapter 1 of Worlds of Awareness: Cetaceans, Evolution, and Cultures of Consciousness.

I’m looking for critical readers willing to engage with full chapters — particularly people with backgrounds in philosophy of mind, history and philosophy of science, or intellectual history, but also thoughtful readers of any background who can tell me where the argument lost them. If you’re interested, please contact me: rsm at 137fsc dot net.